

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I _Pusalapati Chandu_(192372119) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *Pchandu*

Question 1: Event Communication Security

Scenario

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, the message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

Introduction

The Caesar Cipher is one of the oldest and simplest classical encryption techniques. It replaces each letter of the plaintext with another letter that is a fixed number of positions ahead in the alphabet. In this scenario, a shift value of 4 is used to protect the message before it is shared with authorized members.

Theory of Caesar Cipher

Plain Letter	Cipher Letter (Shift +4)
A	E
B	F
C	G
D	H
E	I
M	Q
N	R
O	S
T	X
W	A
X	B
Y	C
Z	D

(a) Encryption Process

Plaintext: MEET AT NOON

Shift Value: +4

Step-by-step conversion:

M→Q

E→I

E→I

T→X

A→E

T→X

N→R

O→S

O→S

N→R

Encrypted Message (Ciphertext):

QIIX EX RSSR

Encryption Formula:

$$C = (P + K) \bmod 26$$

Where C = Ciphertext, P = Plaintext, K = Shift Value (4).

(b) Decryption Process

An authorized member who knows the shift value of 4 decrypts the message by shifting every encrypted letter backward by four positions.

Cipher Letter	Recovered Plain Letter
Q	M
I	E
I	E
X	T
E	A
X	T
R	N
S	O
S	O
R	N

Recovered Message: MEET AT NOON

Decryption Formula:

$$P = (C - K) \bmod 26$$

Advantages

- Easy to understand and implement.
- Fast encryption and decryption.
- Useful for learning cryptography fundamentals.
- Requires minimal computational resources.

Limitations

- Only 25 possible keys, making brute-force attacks easy.
- Vulnerable to frequency analysis.
- Not suitable for protecting modern confidential data.

Applications

- Educational demonstrations of encryption.
- Puzzle creation and coding games.
- Learning the fundamentals of cryptography.
- Historical military communication.

Conclusion

Using a Caesar Cipher with a shift of 4, the plaintext "MEET AT NOON" is encrypted as "QIIX EX RSSR". An authorized receiver can recover the original message by shifting each ciphertext letter backward by four positions. Although this cipher demonstrates the basic principles of symmetric encryption, it is not secure enough for modern communication.

Question 2: Intercepted Suspicious Message

Scenario

A cybersecurity intern intercepts the coded message "GSRH RH Z HVXIVG" during network monitoring. The message appears to use a simple substitution cipher rather than a modern encryption algorithm.

Introduction

A substitution cipher replaces each letter of the plaintext with another letter according to a fixed rule. One well-known classical substitution cipher is the Atbash cipher, which reverses the alphabet ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, etc.). Because every letter always maps to the same corresponding letter, it is easy to analyze using letter patterns.

(a) Deduce the Original Message

Ciphertext: GSRH RH Z HVXIVG

Using the Atbash cipher:

Cipher Letter	Decoded Letter	Explanation
G	T	$G \leftrightarrow T$
S	H	$S \leftrightarrow H$
R	I	$R \leftrightarrow I$
H	S	$H \leftrightarrow S$
Z	A	$Z \leftrightarrow A$
H	S	$H \leftrightarrow S$
V	E	$V \leftrightarrow E$
X	C	$X \leftrightarrow C$
I	R	$I \leftrightarrow R$
V	E	$V \leftrightarrow E$
G	T	$G \leftrightarrow T$

Recovered Plaintext:

THIS IS A SECRET

(b) Justification

1. The repeated word "RH" appears twice and decodes to "IS", indicating a consistent substitution.
2. The single-letter word "Z" becomes "A", matching common English sentence structure.
3. The final word "HVXIVG" decodes to "SECRET", which forms a meaningful sentence.
4. The alphabet reversal ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$...) exactly matches the Atbash cipher.
5. The message contains readable English after decoding, confirming the correct cipher.

Characteristics of the Atbash Cipher

- A monoalphabetic substitution cipher.
- Uses a fixed reversed alphabet.
- No encryption key is required.
- Easy to implement but insecure.
- Vulnerable to frequency analysis and pattern matching.

Advantages

- Simple to understand and use.
- Fast encryption and decryption.
- Useful for teaching classical cryptography.

Limitations

- No secret key for security.
- Can be broken easily using frequency analysis.
- Unsuitable for modern confidential communication.

- **Applications**

- Educational Purposes

- Used to teach students the fundamentals of substitution ciphers and classical cryptography.

- Cryptography Learning

- Helps beginners understand encryption, decryption, and pattern analysis before learning advanced algorithms like AES and RSA.

- Puzzle and Brain Games

- Frequently used in crossword puzzles, escape rooms, treasure hunts, and cryptographic challenges.

- Historical Studies

- Used by historians and researchers to study ancient encrypted texts and early communication methods.

- Cybersecurity Training

- Used in cybersecurity courses and Capture The Flag (CTF) competitions to practice cryptanalysis and recognize simple substitution ciphers.

Conclusion

The intercepted ciphertext "GSRH RH Z HVXIVG" is most likely encrypted using the Atbash substitution cipher. Decoding it reveals the original message "THIS IS A SECRET". Pattern analysis, repeated words, and the reversed alphabet confirm the correctness of the solution. Although historically important, the Atbash cipher does not provide adequate security for modern communication.

Question 3: Secure Messaging System

Scenario

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

(a) Step-by-Step Encryption Using the Vigenère Cipher

Step 1: Plaintext

HELLO

Step 2: Keyword

KEY

Repeated keyword: KEYKE

Plaintext	Value	Keyword	Value	Ciphertext
H	7	K	10	R
E	4	E	4	I
L	11	Y	24	J
L	11	K	10	V
O	14	E	4	S

Encryption Formula:

Ciphertext = (Plaintext + Keyword) mod 26

Final Ciphertext: RIJVS

(b) Effect of the Repeating Keyword

The keyword KEY is repeated as KEYKE so that each plaintext letter has a corresponding key letter. Each key letter produces a different shift value:

1. $H + K = (7 + 10) \bmod 26 = 17 = R$
2. $E + E = (4 + 4) \bmod 26 = 8 = I$
3. $L + Y = (11 + 24) \bmod 26 = 35 \bmod 26 = 9 = J$
4. $L + K = (11 + 10) \bmod 26 = 21 = V$
5. $O + E = (14 + 4) \bmod 26 = 18 = S$

$H + K = R$

$E + E = I$

$L + Y = J$

$L + K = V$

$O + E = S$

Unlike the Caesar cipher, the shift changes according to the keyword. This makes the ciphertext more difficult to analyze because identical plaintext letters can produce different ciphertext letters depending on their position.

Advantages

- More secure than the Caesar cipher.
- Uses a secret keyword for encryption and decryption.
- Reduces the effectiveness of simple frequency analysis.
- Easy to understand and implement.

Limitations

- Short or repeated keywords are vulnerable to cryptanalysis.
- Not suitable for modern secure communication.
- Weaker than modern algorithms such as AES.

Conclusion

Using the Vigenère cipher, the plaintext "HELLO" is encrypted with the keyword "KEY" to produce the ciphertext "RIJVS". The repeating keyword changes the shift for each character, making the encryption stronger than a simple substitution cipher.

Question 4: Suspicious Digital Asset Investigation

Scenario

During a digital forensic audit, investigators suspect that a company logo image may contain hidden confidential information. Although the image appears normal, it may have secret data embedded using steganography.

Introduction

Steganography is the practice of hiding secret information inside digital media such as images, audio, or video. Unlike encryption, which hides the content of a message, steganography hides the existence of the message itself. Digital forensic investigators use specialized techniques to detect and analyze hidden information.

(a) Detecting Hidden Data in the Image

Investigators can follow these steps:

1. Visual Inspection: Compare the suspected image with the original for unusual artifacts.
2. Metadata Analysis: Examine EXIF and file metadata for unexpected information.
3. File Size Comparison: A larger-than-expected file size may indicate embedded data.
4. Histogram Analysis: Compare pixel distribution to identify abnormal changes.
5. LSB (Least Significant Bit) Analysis: Check whether the least significant bits of pixels have been modified to hide data.
6. Signature Verification: Verify the file header and image format for inconsistencies.
7. Use Steganalysis Tools: Tools such as StegExpose, zsteg, StegSolve, or forensic suites can help detect hidden content.

Example Investigation Flow



(b) Techniques to Strengthen Hidden Communications

Technique 1: Encrypt the message before embedding it.

Even if hidden data is extracted, strong encryption (such as AES) prevents attackers from reading the confidential information without the decryption key.

Technique 2: Use Randomized or Key-Based Steganography.

Instead of embedding data in predictable pixel locations, use a secret key to determine where data is hidden. This makes steganalysis and extraction much more difficult.

Advantages

- Protects confidential information from casual observation.
- Combines cryptography and steganography for stronger security.
- Useful in digital forensics and secure communication.

Limitations

- Advanced forensic tools can detect poorly implemented steganography.
- Image compression or editing may damage hidden data.
- Steganography alone does not protect message contents without encryption.

Conclusion

Investigators can detect hidden information through metadata analysis, histogram inspection, LSB analysis, and specialized steganalysis tools. To improve security, organizations should encrypt the secret message before embedding it and use key-based randomized embedding techniques to resist detection and extraction.

Question 5: Legacy Encryption in Corporate Communication

Scenario

A legacy system uses a transposition-based encryption method with three rails. The received ciphertext is:

WECRLTEERDSOEFEAOCAVDEN

Introduction

The Rail Fence Cipher is a classical transposition cipher. Unlike substitution ciphers, it does not change the letters of the message; instead, it rearranges their positions. In a 3-rail Rail Fence cipher, the plaintext is written diagonally across three rows and then read row by row to produce the ciphertext.

(a) Reverse the Encryption Process

Ciphertext:

WECRLTEERDSOEFEAOCAVDEN

Assuming a 3-rail Rail Fence cipher, reconstruct the zigzag pattern and read the letters in the original order.

Recovered Plaintext:

WE ARE DISCOVERED FLEE AT ONCE

Rail Fence Reconstruction

Rail 1 : W . . . E . . . C . . . R . . . L . . . T . . . E

Rail 2 : . E . R . D . S . O . E . E . F . E . A . O . C .

Rail 3 : . . A . . . I . . . V . . . D . . . E . . . N

Reading diagonally reconstructs:

WE ARE DISCOVERED FLEE AT ONCE

(b) Why is this Method Vulnerable?

- Only the positions of characters change; the letters themselves remain unchanged.
- Letter frequency is preserved, making cryptanalysis easier.
- Attackers can quickly try different rail numbers using brute force.
- No secret cryptographic key is used.
- Modern computers can break this cipher within seconds.

(c) Practical Enhancement

Replace the Rail Fence cipher with a modern encryption algorithm such as AES-256. An additional improvement is to encrypt the message first and then, if required for legacy compatibility, apply the transposition cipher as a second layer. This ensures that even if the transposition is reversed, the message remains protected without the AES key.

Advantages of the Enhancement

- Provides strong confidentiality.
- Resists brute-force and statistical attacks.
- Widely accepted for secure business communication.
- Supports secure key management.

Conclusion

The ciphertext 'WECRLTEERDSOEFEAOCAIVDEN' decrypts to 'WE ARE DISCOVERED FLEE AT ONCE' using the 3-rail Rail Fence cipher. Although historically significant, this cipher is insecure because it can be broken using modern cryptanalysis. Replacing it with AES-based encryption greatly improves confidentiality.